

Handling Missing Data in ETL

Assignment



Objective

This DPP helps understand:

- Why missing data occurs in ETL pipelines
- How different handling techniques impact analytics
- How to **choose the right method** instead of blindly deleting data

SECTION A – THEORETICAL QUESTIONS

Q1. What are the **most common reasons** for missing data in ETL pipelines?

Q2. Why is **blindly deleting rows with missing values** considered a bad practice in ETL?

Q3. Explain the difference between:

- **Listwise deletion**
- **Column deletion**

Also mention **one scenario** where each is appropriate.

Q4. Why is **median imputation** preferred over mean imputation for skewed data such as income?

Q5. What is **forward fill** and in what type of dataset is it most useful?

Q6. Why should **flagging missing values** be done before imputation in an ETL workflow?

Q7. Consider a scenario where income is missing for many customers.

How can this missingness itself provide **business insights**?

SECTION B – PRACTICAL QUESTIONS

Customer_ID	Name	City	Monthly_Sales	Income	Region
101	Rahul Mehta	Mumbai	12000	65000	West
102	Anjali Rao	Bengaluru	NaN	NaN	South
103	Suresh Iyer	Chennai	15000	72000	South
104	Neha Singh	Delhi	NaN	NaN	North
105	Amit Verma	Pune	18000	58000	NaN
106	Karan Shah	Ahmedabad	NaN	61000	West
107	Pooja Das	Kolkata	14000	NaN	East
108	Riya Kapoor	Jaipur	16000	69000	North

Use the given dataset for all questions.

Q8. Listwise Deletion

Remove all rows where **Region** is missing.

Tasks:

1. Identify affected rows
2. Show the dataset after deletion
3. Mention how many records were lost

Q9. Imputation

Handle missing values in **Monthly_Sales** using:

- **Forward Fill**

Tasks:

1. Apply forward fill
2. Show before vs after values
3. Explain why forward fill is suitable here

Q10. Flagging Missing Data

Create a **flag column** for missing Income.

Tasks:

1. Create Income_Missing_Flag (0 = present, 1 = missing)
2. Show updated dataset
3. Count how many customers have missing income

